# KIGALI COLLEGE

- Option: INFORMATION TECHNOLOGY
- Module: BACKEND DEVELOPMENT USING JAVA
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- Class: Y2 IT

. Dates: 22 Feb, 2026

# YOUTH TALENT SHOWCASE & OPPORTUNITY PORTAL

**ABSTRACT**

The Youth Talent Showcase & Opportunity Portal is a Java-based web application developed using JSP, Servlets, JDBC, and MySQL following the MVC architecture. The system allows young individuals to showcase their talents, receive ratings and comments, and gain visibility. The platform includes user authentication, role-based access control, admin approval workflow, rating system, comment management, and badge features. The system ensures proper validation, security, and database normalization up to Third Normal Form (3NF).

# 1. INTRODUCTION

This project was developed to provide a digital platform where youth can present their talents and connect with opportunities. The system solves the problem of limited exposure for talented youth by providing an online management and evaluation system.

# 2. PROJECT PROPOSAL

## 2.1 Problem Statement

Many talented youths lack a structured platform to showcase their skills and receive recognition. There is a need for a centralized web-based system to manage talent submissions, ratings, and approvals.

## 2.2 Objectives

**General Objective:**
To design and develop a web-based portal that allows youth to showcase talents and connect with opportunities.
Specific Objectives:

- Implement user authentication system
- Develop CRUD operations for talents
- Implement rating and comment system
- Create admin approval workflow
- Ensure secure database integration

## 2.3 Scope of the System

The system includes user registration, login, talent submission, approval workflow, rating system, comment management, search and filtering features, and administrative dashboard.

## 2.4 Technologies Used

- Java (JDK 8+)
- JSP & Servlets
- MySQL Database
- JDBC
- Apache Tomcat Server

- Bootstrap for UI

### 2.5 Expected Outcomes

A fully functional MVC-based web application with complete CRUD operations, authentication system, role-based access control, and normalized database.

## 3. SYSTEM ANALYSIS

### 3.1 Functional Requirements

- User Registration and Login
- Talent CRUD operations
- Rating and Comment system
- Admin approval/rejection
- Search and filtering

### 3.2 Non-Functional Requirements

- System security
- Performance efficiency
- Usability and responsive design
- Data integrity and consistency

### 3.3 User Roles

- Admin
- Registered User
- Public Visitor

## 4. SYSTEM DESIGN

- Architecture: MVC (Model-View-Controller)
- Model: JavaBeans & DAO classes
- View: JSP Pages
- Controller: Servlets

## 5. DATABASE DESIGN

- Database Name: youth_talent_portal
- Total Tables: 8
- Primary Keys and Foreign Keys implemented
- Database normalized up to Third Normal Form (3NF)

### Sample SQL Scripts

```
CREATE DATABASE youth_talent_portal;
CREATE TABLE users (id INT PRIMARY KEY AUTO_INCREMENT, username VARCHAR(50),
password VARCHAR(255), role VARCHAR(20));
```

## 6. APPLICATION IMPLEMENTATION

- Model Layer: JavaBeans
- DAO Layer: Database access classes
- Controller Layer: Servlets
- View Layer: JSP Pages
- JDBC Connection class implemented

## 7. BUSINESS LOGIC IMPLEMENTATION

- Server-side validation
- Status changes (Pending → Approved)
- Business rules enforcement
- Error handling and messages

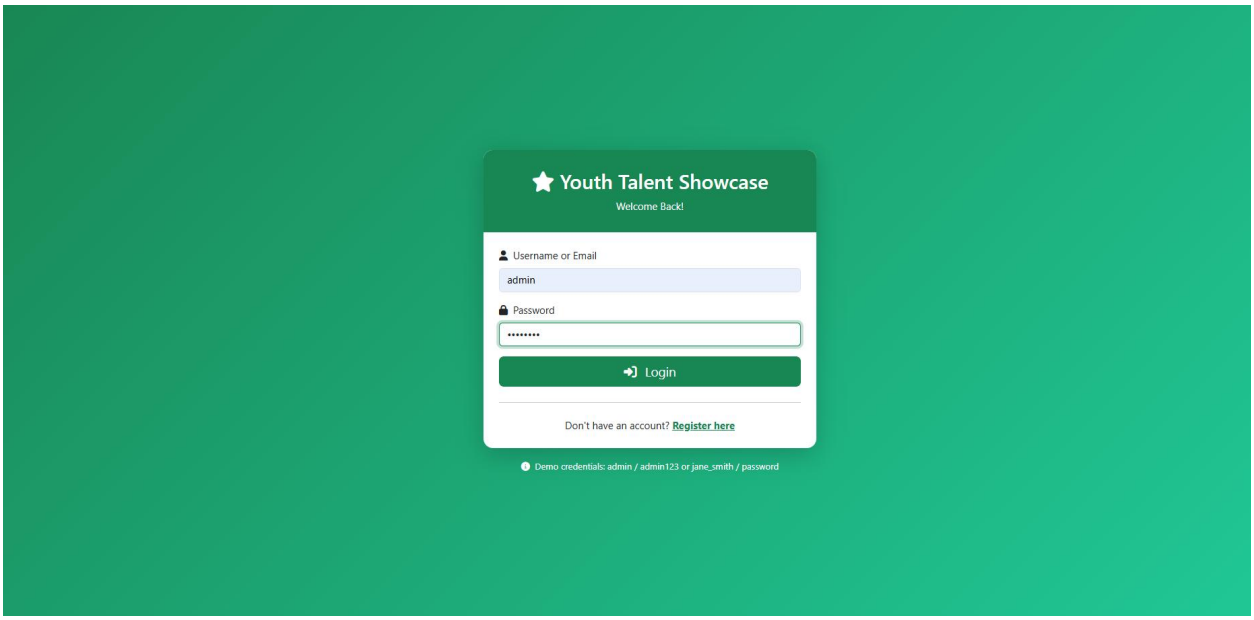
## 8. USER INTERFACE

Bootstrap-based responsive design
Forms, Tables, and Navigation menus implemented

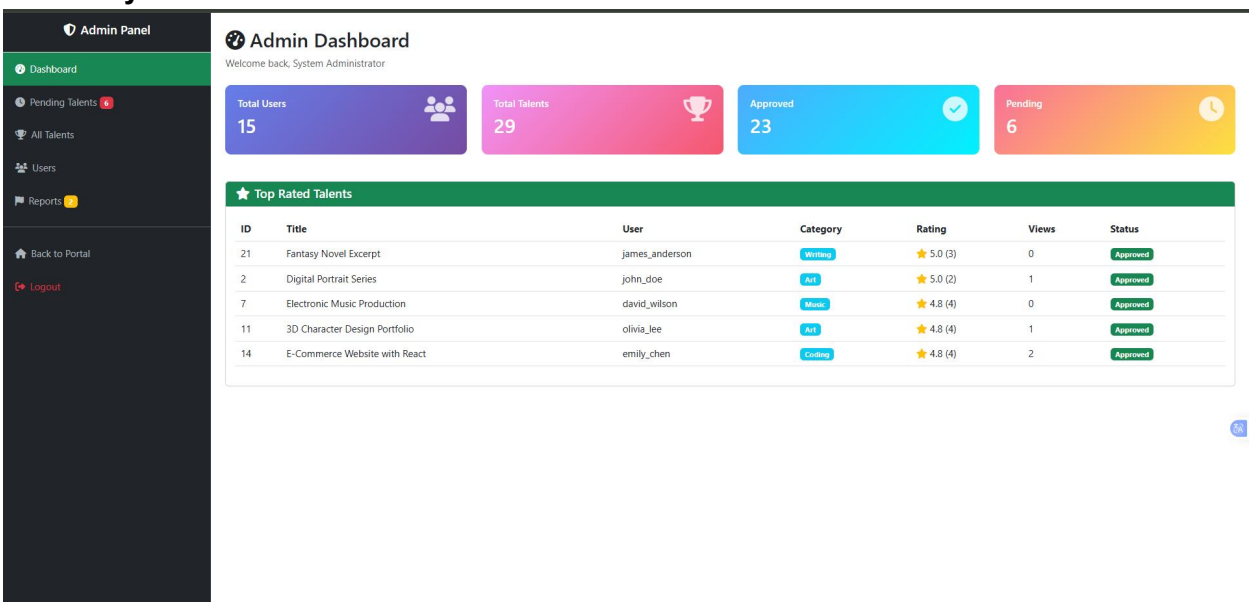
## 9. TESTING & RESULTS

60 test cases executed successfully
**Authentication testing**

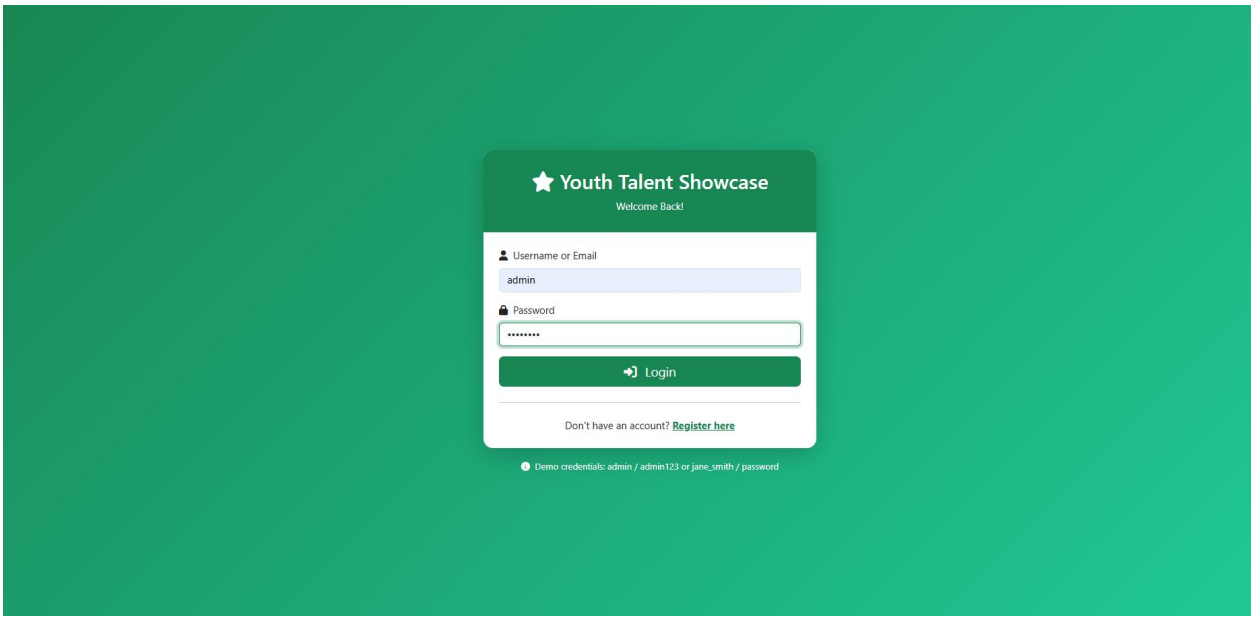
- **Try login with:admin**

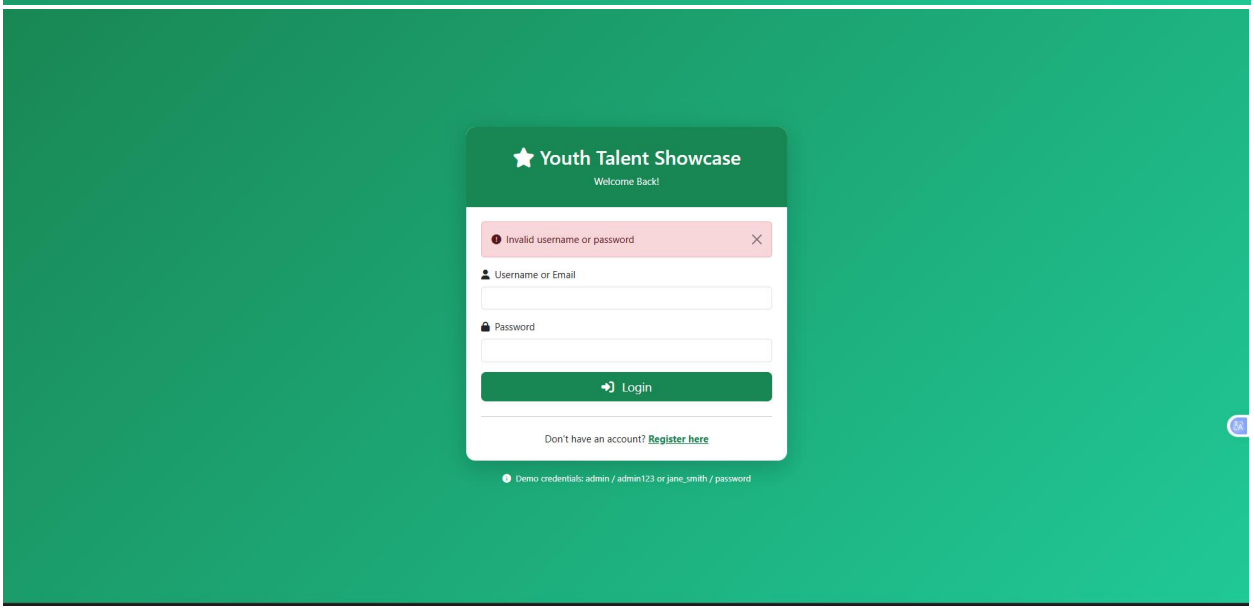



- **Verify redirect to dashboard**




- **Try login with wrong password => should show an error**




**CRUD operation testing**

**Add Talent(So far this user has three talent as the status shown)**

# My Talents

Manage and track all your talent submissions

| 3 | 2 | 1 | 0 |
|---|---|---|---|
| Total Talents | Approved | Pending | Rejected |

+ Add New Talent | Explore All Talents

| Image | Title | Category | Status | Rating | Views | Posted | Actions |
|---|---|---|---|---|---|---|---|
| | **AI-Powered Study Assistant**<br>A web application that helps students organize the... | Coding | APPROVED | 4.5 (2) | 0 | Feb 20, 2026 | |
| | **Smart Recycling System**<br>An IoT-based system that automatically sorts recyc... | Innovation | PENDING | 0.0 (0) | 0 | Feb 20, 2026 | |
| | **Smart Home Automation System**<br>An IoT-based home automation system controllable v... | Innovation | APPROVED | 4.8 (4) | 0 | Feb 20, 2026 | |

Previous 1 2 Next

Youth Talent Portal | Dashboard | Explore Talents | My Talents | Add Talent | jane_smith

# Add New Talent

Showcase your amazing skills and creativity

**Talent Title** *

Enter a catchy title for your talent

Choose a clear, descriptive title (5-200 characters) 0/200

**Category** *

Select a category...

Select the category that best describes your talent

**Description** *

Describe your talent in detail. What makes it special? What inspired you?

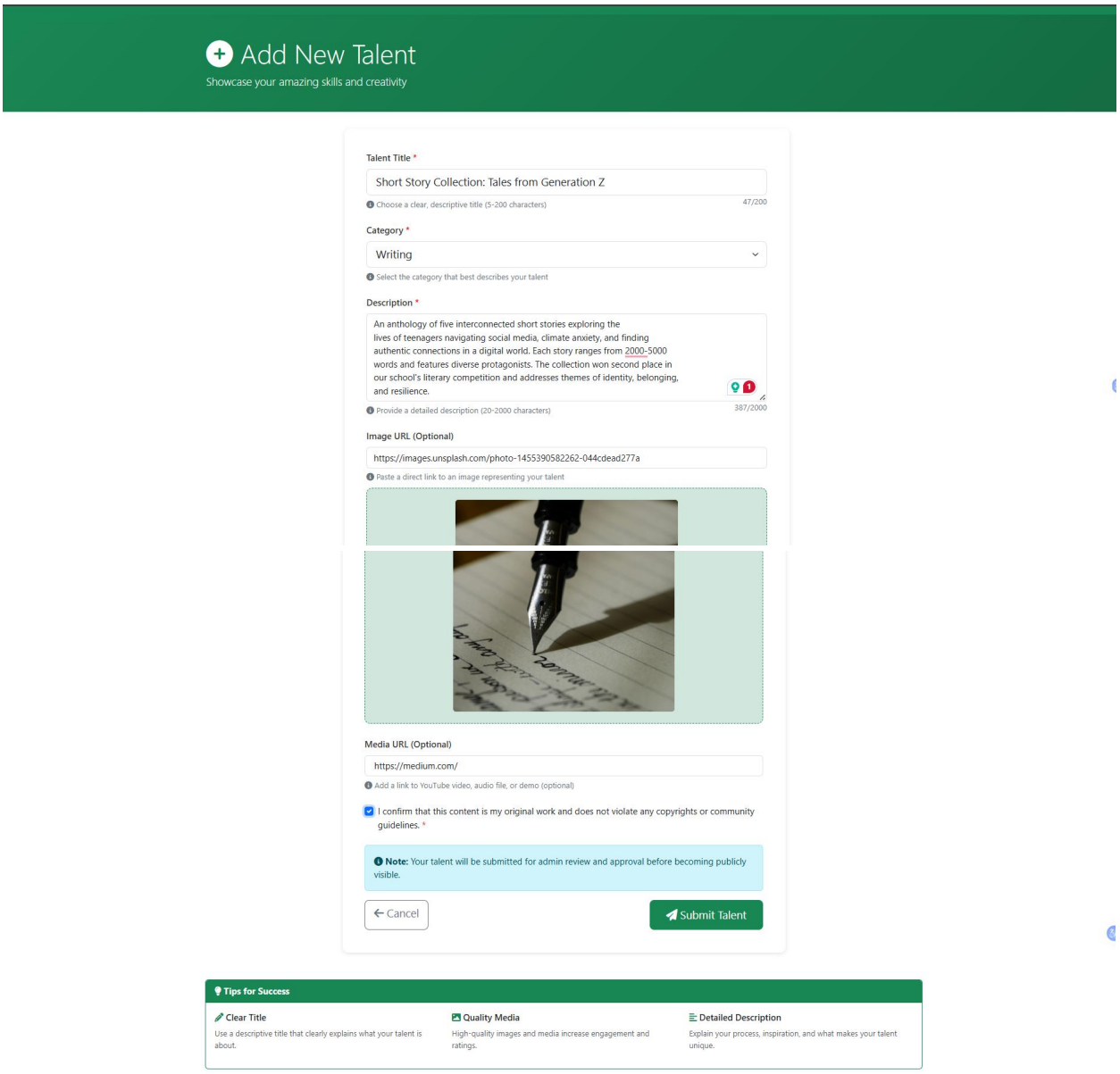
Provide a detailed description (20-2000 characters) 0/2000

**Image URL (Optional)**

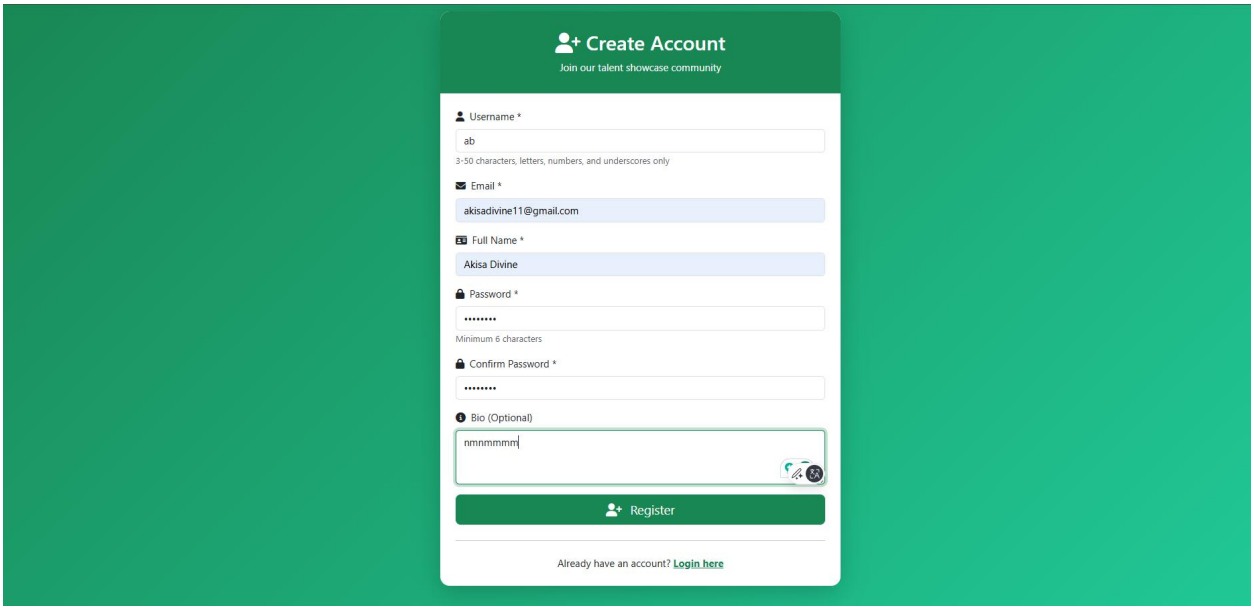
https://example.com/image.jpg

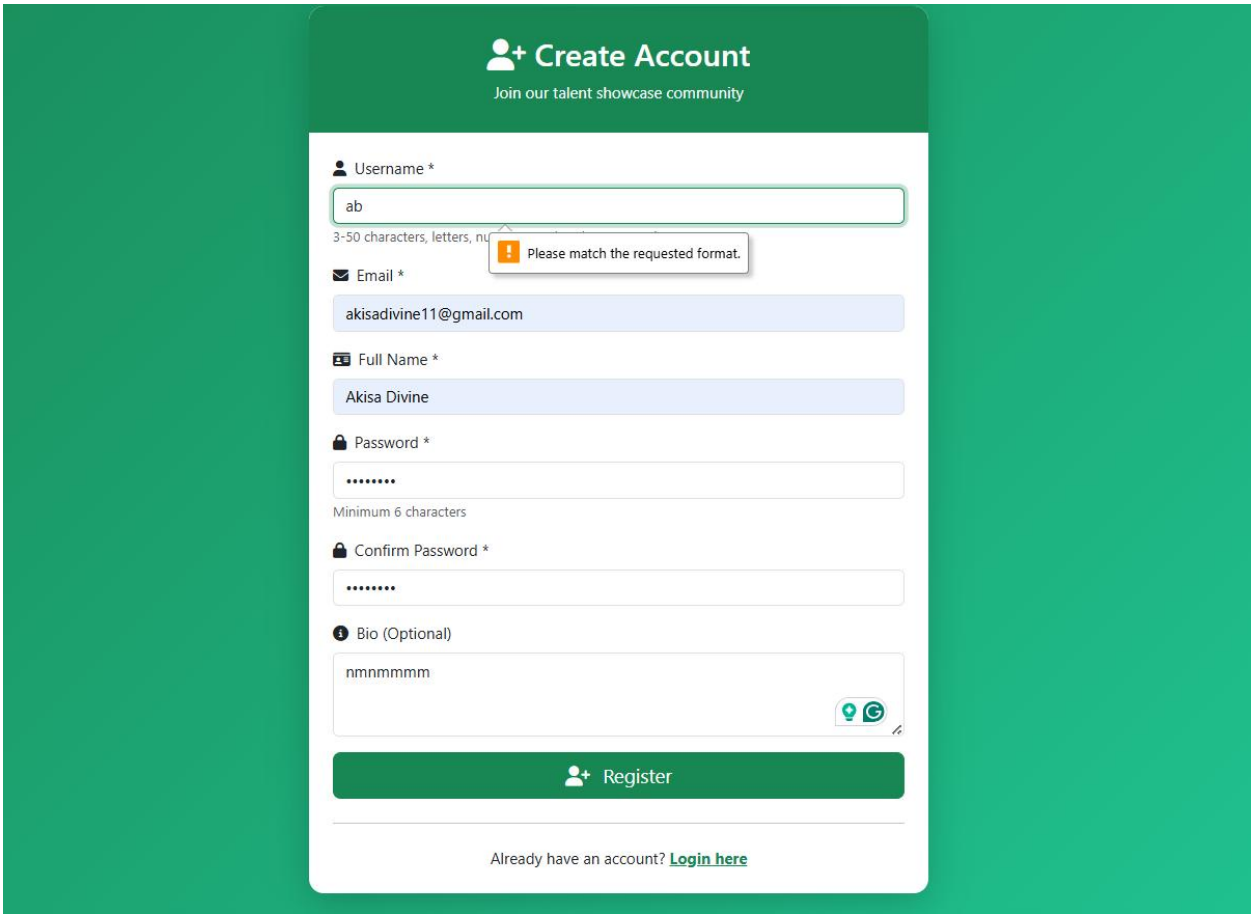
Paste a direct link to an image representing your talent

**Media URL (Optional)**

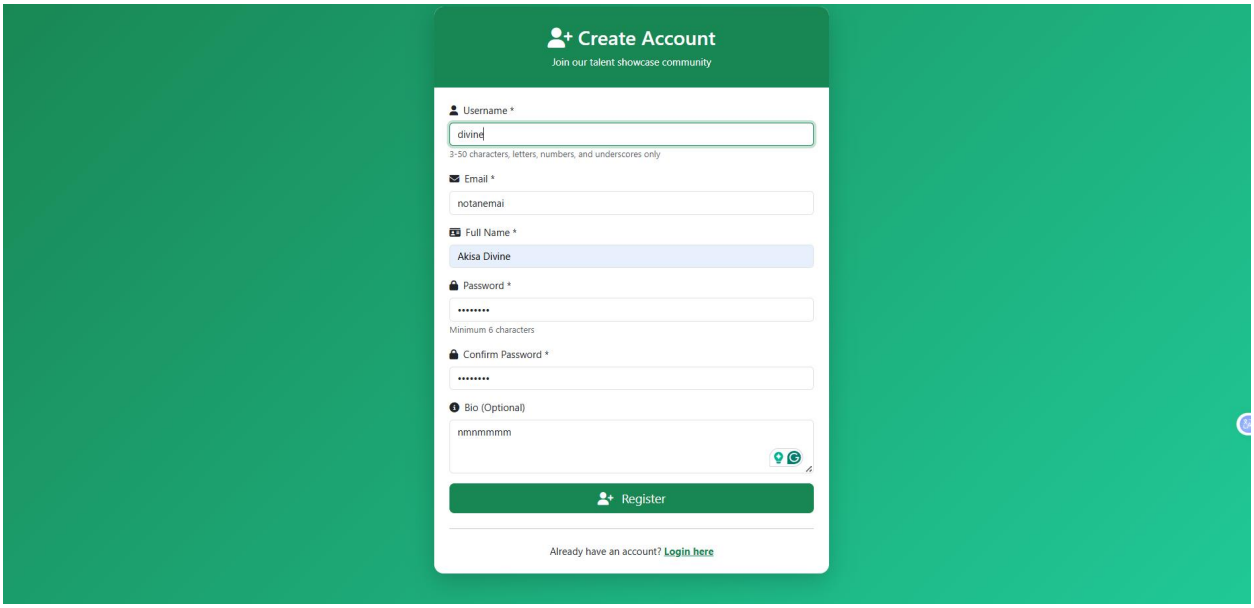


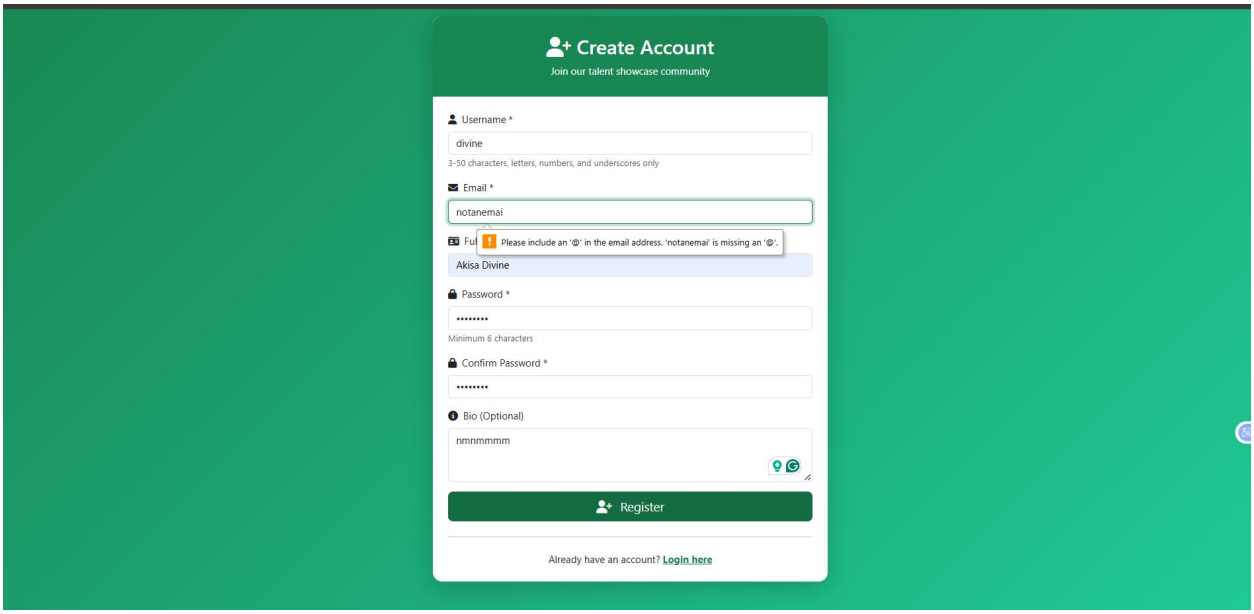

**Validation error testing**
**Try registering with username ab=> should fail(too short)**



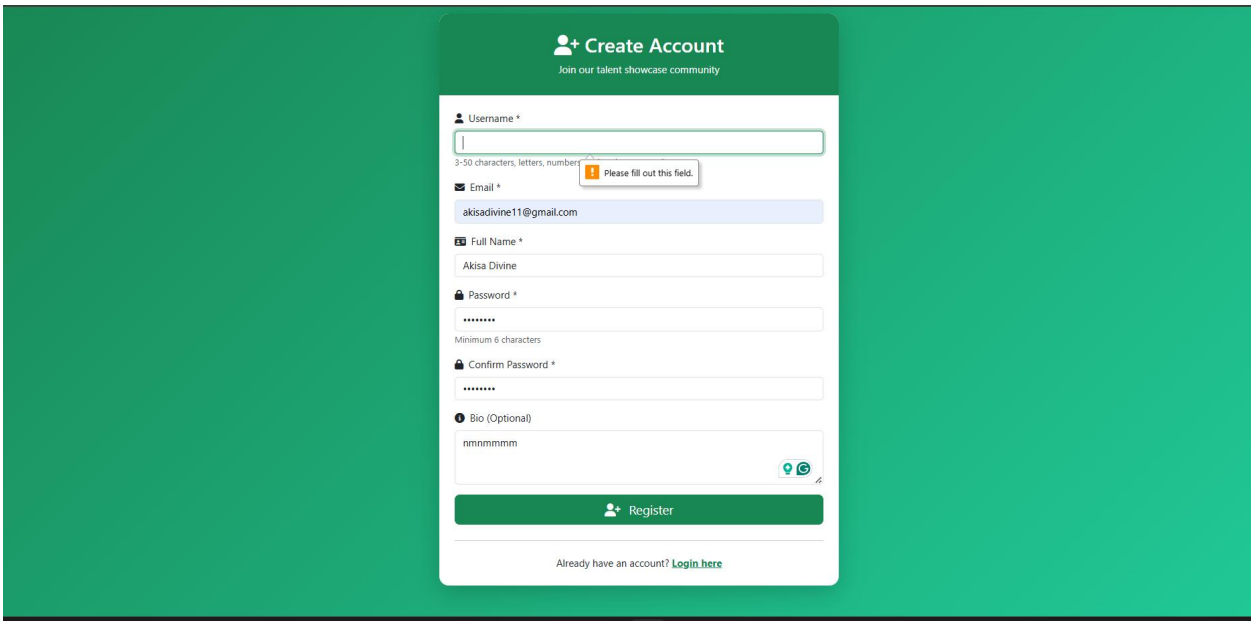




**Try email ‘’notanemail” => should fail**

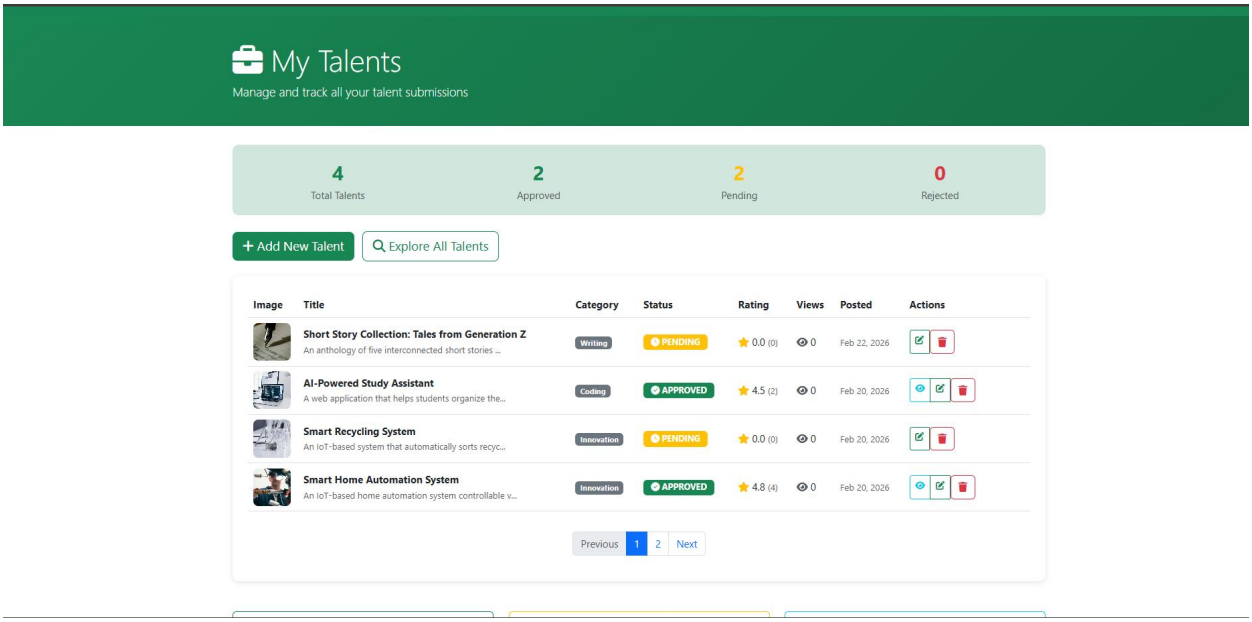


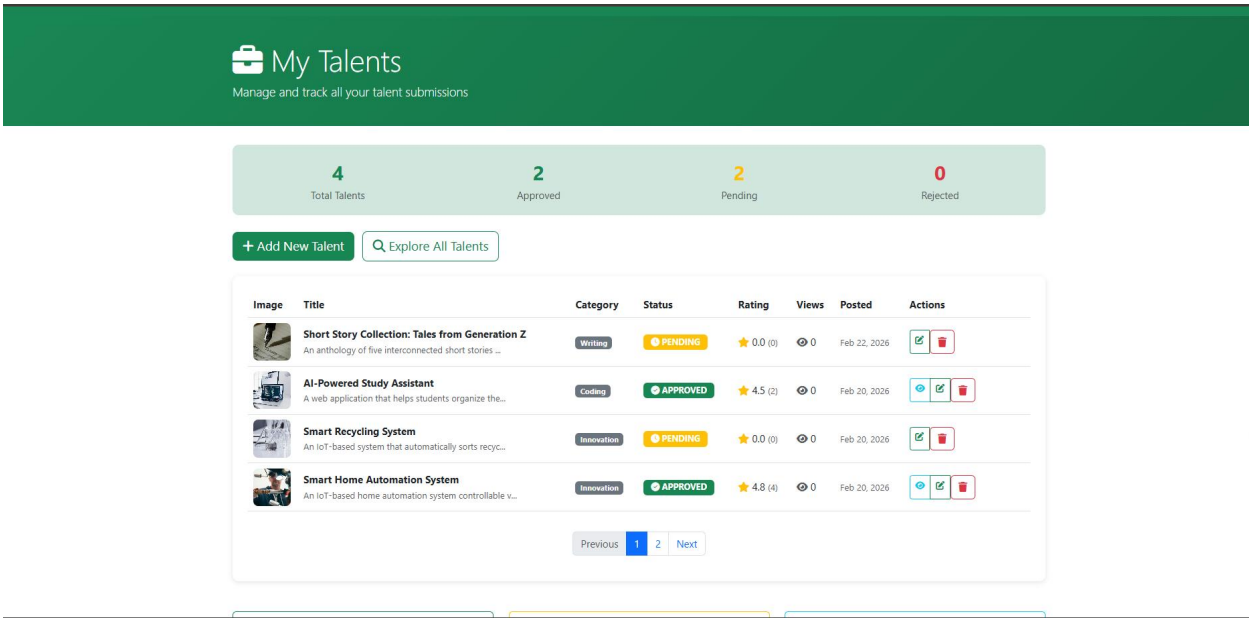

**Try empty fields => should show required errors**



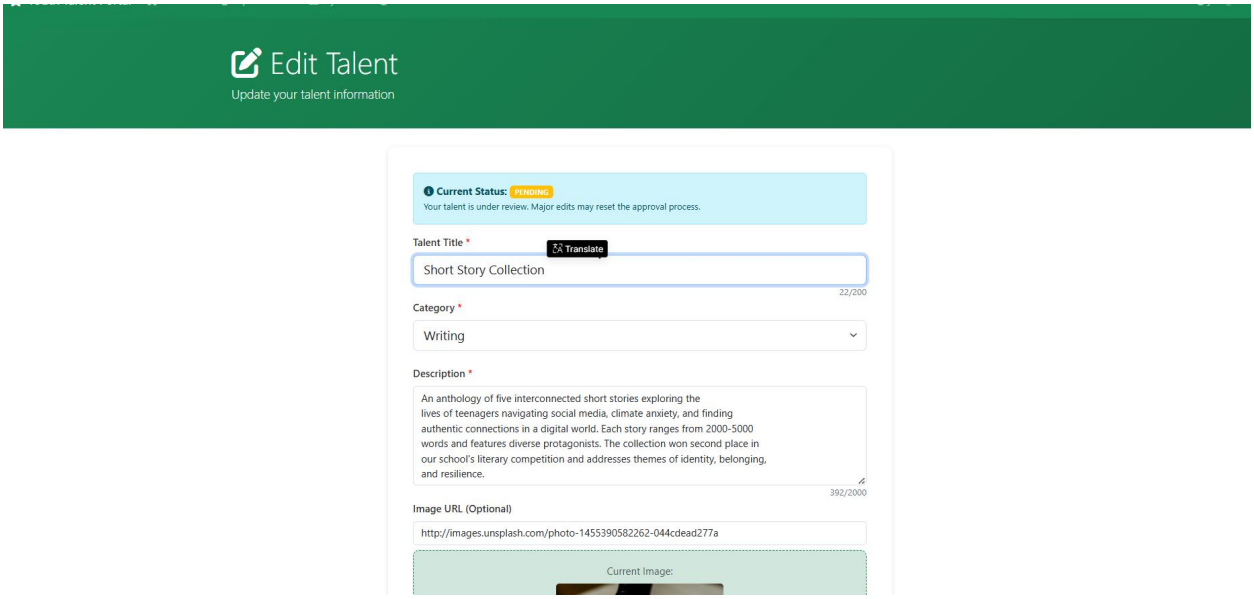

**Now our short story talent added and the status is pending(Select all talent)**

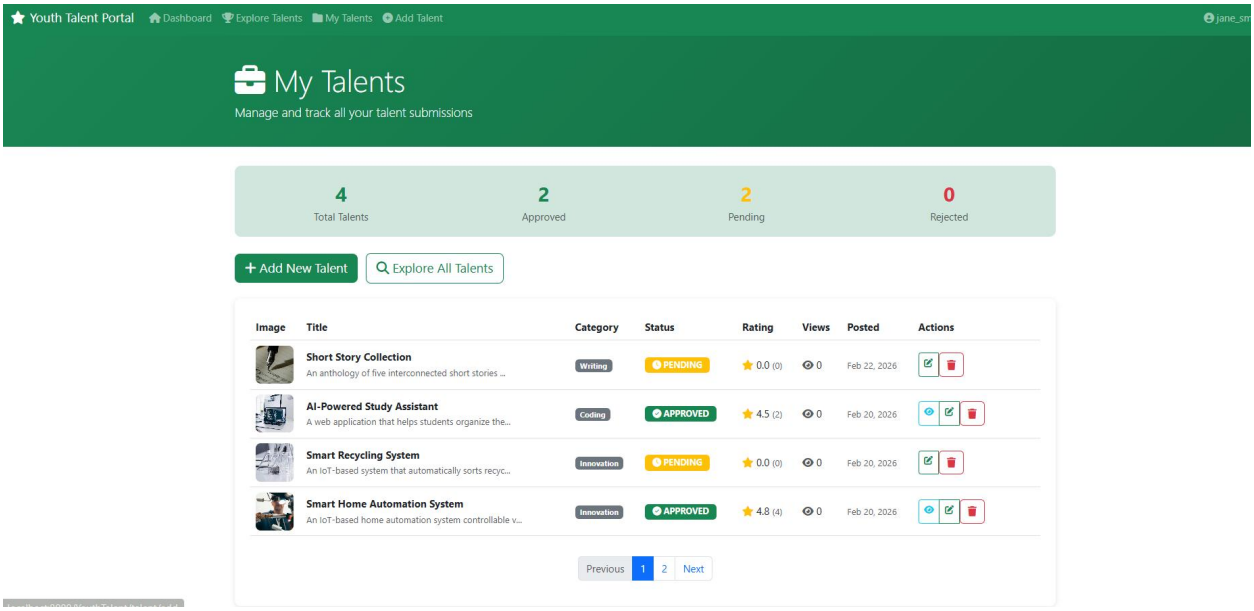



**Update the talent  from short story collection: tales from generation z to short story collection only**
**before update**



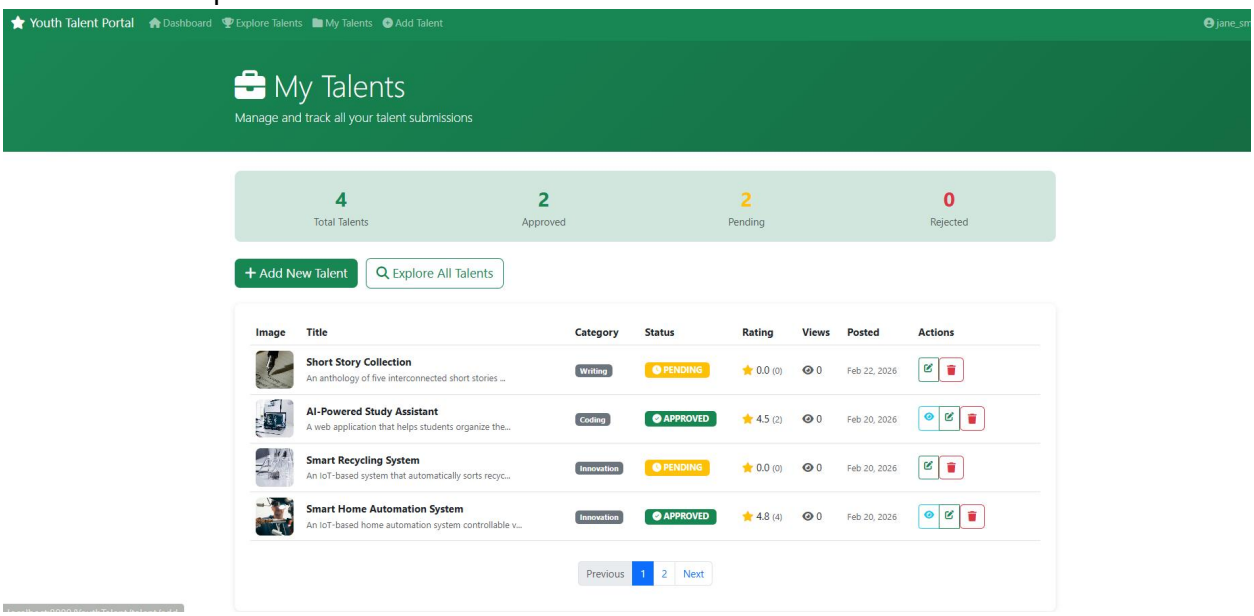

After Update



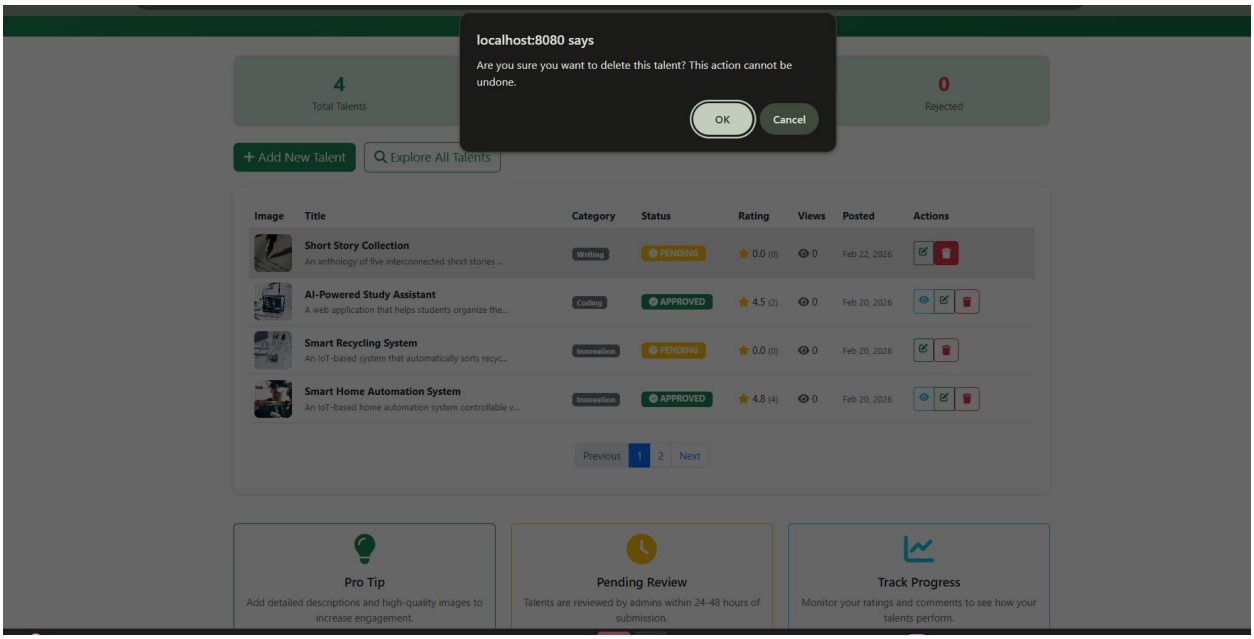


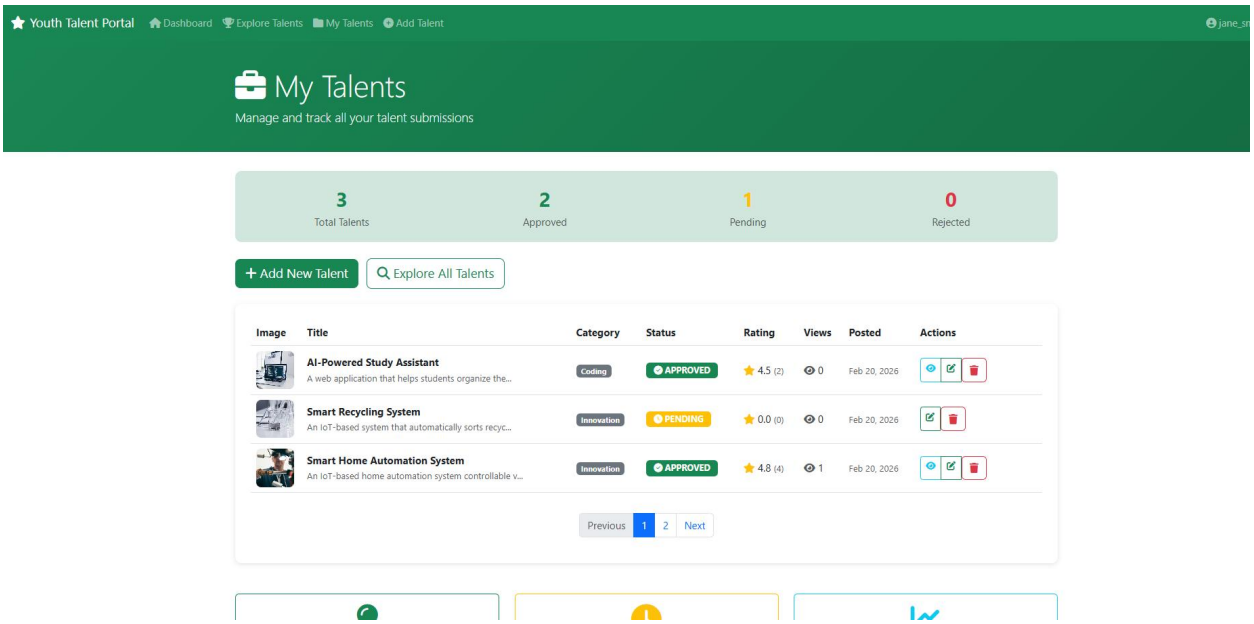

Delete Short Story Collection

Before Delete Operation




After Delete

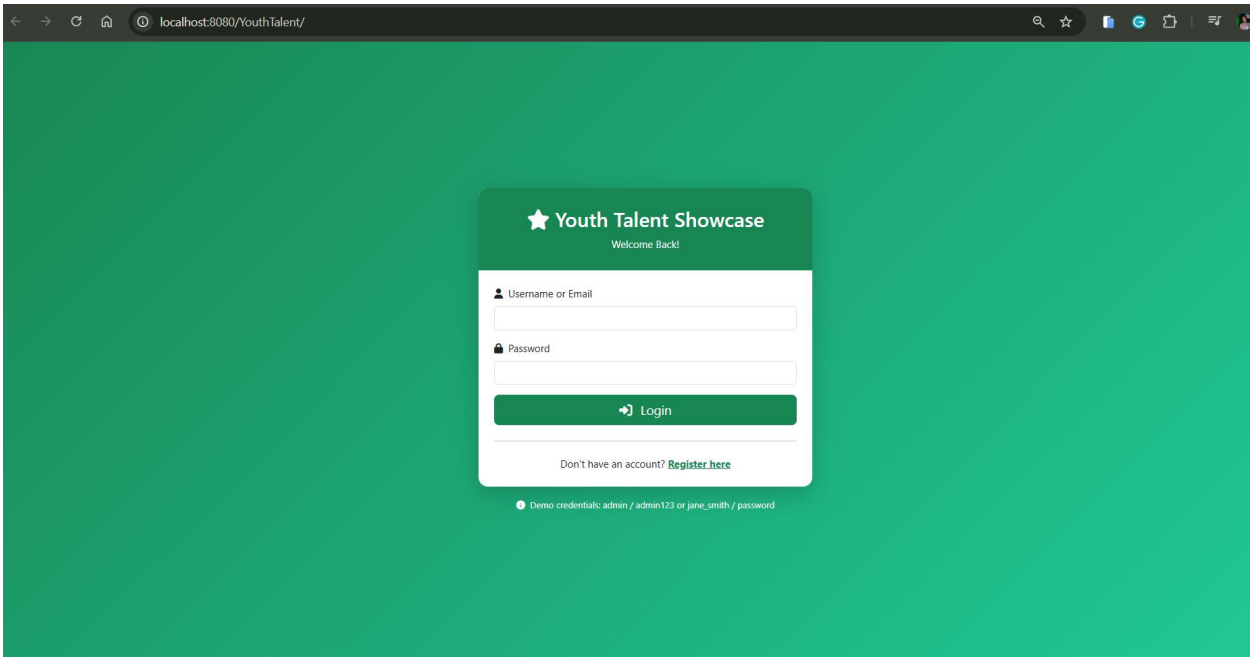


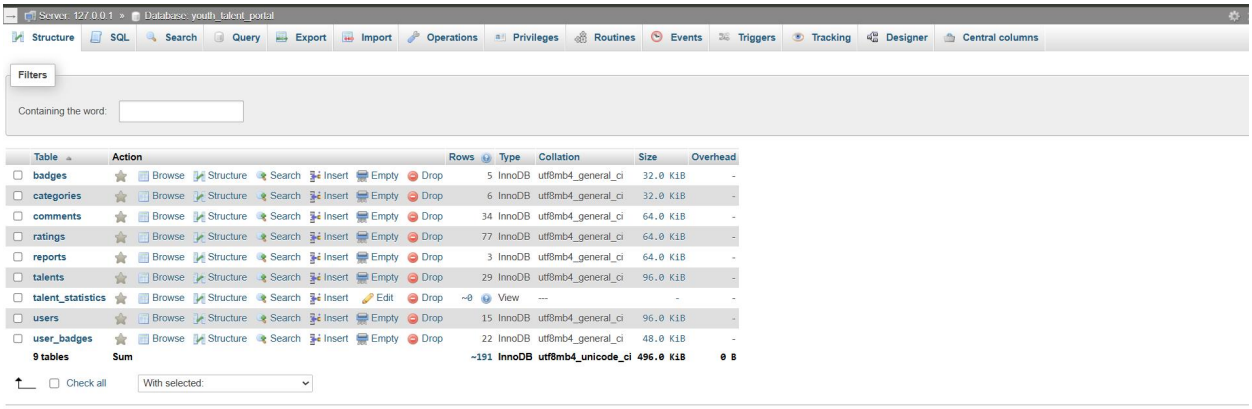

# 10. DEPLOYMENT REQUIREMENTS

- Apache Tomcat Server

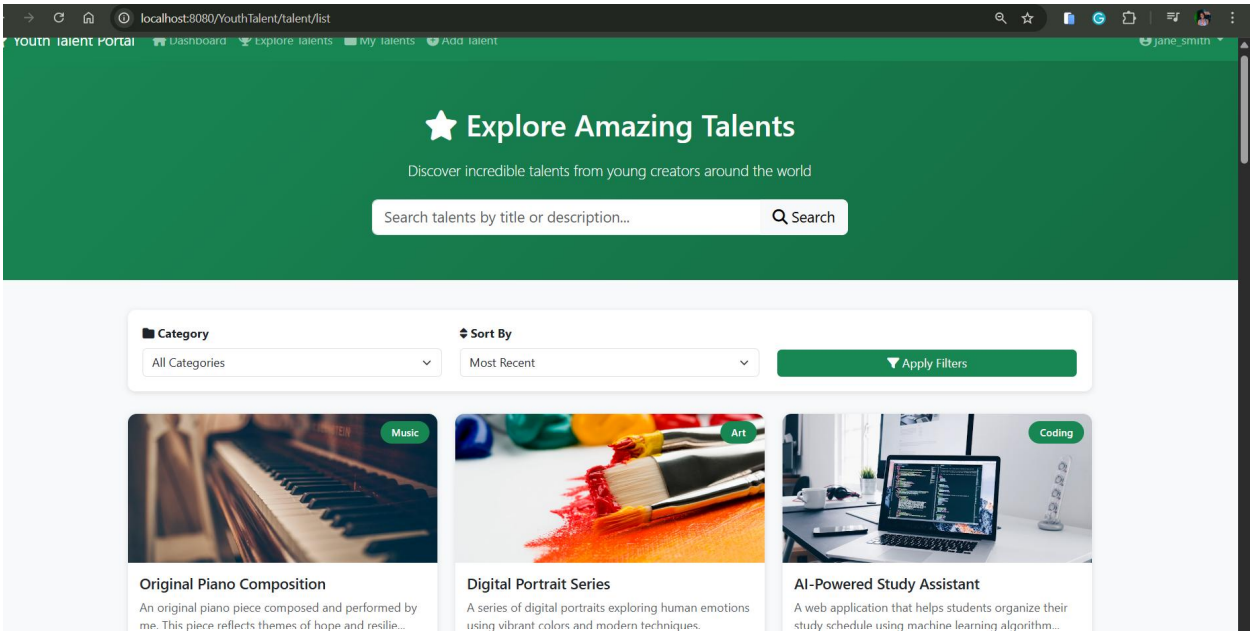
```
Tomcat
22-Feb-2026 19:36:02.557 WARNING [localhost-startStop-1] org.apache.tomcat.util.descriptor.web.WebXml.setVersion Unknown version string [4.0]. Default versi
on will be used.
22-Feb-2026 19:36:03.547 INFO [localhost-startStop-1] org.apache.jasper.servlet.TldScanner.scanJars At least one JAR was scanned for TLDs yet contained no T
LDs. Enable debug logging for this logger for a complete list of JARs that were scanned but no TLDs were found in them. Skipping unneeded JARs during scanni
ng can improve startup time and JSP compilation time.
22-Feb-2026 19:36:03.628 INFO [localhost-startStop-1] org.apache.catalina.startup.HostConfig.deployDirectory Deployment of web application directory [C:\xam
pp\tomcat\webapps\YouthTalent] has finished in [1,095] ms
22-Feb-2026 19:36:03.632 INFO [main] org.apache.coyote.AbstractProtocol.start Starting ProtocolHandler ["http-nio-8080"]
22-Feb-2026 19:36:03.687 INFO [main] org.apache.catalina.startup.Catalina.start Server startup in 3176 ms
MySQL JDBC Driver loaded successfully!
DEBUG: Login attempt for username: jane_smith
DEBUG: Password hash: 5f4dcc3b5aa765d61d8327deb882cf99
DEBUG UserDAO: Attempting authentication for: jane_smith
DEBUG UserDAO: Password hash: 5f4dcc3b5aa765d61d8327deb882cf99
DEBUG UserDAO: User found in database
DEBUG: Authentication result: SUCCESS
DEBUG: User found: jane_smith (Role: USER)
DEBUG: Session created, redirecting...
DEBUG TalentServlet.doPost() - pathInfo: /delete
DEBUG TalentServlet.doPost() - talentId param: 30
DEBUG: Calling deleteTalent()
DEBUG deleteTalent() - METHOD CALLED
DEBUG deleteTalent() - talentId param: 30
DEBUG deleteTalent() - Parsed talentId: 30, userId: 3
DEBUG deleteTalent() - Ownership verified, attempting delete...
Delete talent 30: 1 rows affected
DEBUG deleteTalent() - Delete SUCCESS
DEBUG deleteTalent() - Redirecting to my-talents
```




- MySQL Database configured




- Project deployed without errors




- WAR file generated

# 11. CONCLUSION

The Youth Talent Showcase & Opportunity Portal successfully demonstrates the implementation of Java EE technologies using MVC architecture. The system meets all academic and technical requirements including CRUD operations, authentication, authorization, database normalization, and testing evidence.